

Given:

To implement Linear Regression using Scikit-learn to predict a continuous target variable and evaluate the model using MAE, MSE, and R^2 score.

DATASET: [Click to Download House Price Prediction Dataset](#)

Linear Regression.py

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
df = pd.read_csv(r"..\\DataSets\\Housing.csv")
df = pd.get_dummies(df, drop_first=True)
X = df.drop("price", axis=1)
y = df["price"]
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
model = LinearRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
print("Mean Absolute Error:", mean_absolute_error(y_test, y_pred))
print("Mean Squared Error:", mean_squared_error(y_test, y_pred))
print("R2 Score:", r2_score(y_test, y_pred))
coefficients = pd.DataFrame({
    "Feature": X.columns,
    "Coefficient": model.coef_
})
print("\nRegression Coefficients")
print(coefficients)
plt.figure(figsize=(8, 6))
plt.scatter(y_test, y_pred)
plt.plot(
    [y_test.min(), y_test.max()],
    [y_test.min(), y_test.max()],
    color="red"
)
plt.xlabel("Actual Price")
plt.ylabel("Predicted Price")
plt.title("Actual vs Predicted Price")
plt.show()
```

Output:

Mean Absolute Error: 970043.4039201637

Mean Squared Error: 1754318687330.6643

R2 Score: 0.6529242642153184

Regression Coefficients

	Feature	Coefficient
0	area	2.359688e+02
1	bedrooms	7.677870e+04
2	bathrooms	1.094445e+06
3	stories	4.074766e+05
4	parking	2.248419e+05
5	mainroad_yes	3.679199e+05
6	guestroom_yes	2.316100e+05
7	basement_yes	3.902512e+05
8	hotwaterheating_yes	6.846499e+05
9	airconditioning_yes	7.914267e+05
10	prefarea_yes	6.298906e+05
11	furnishingstatus_semi-furnished	-1.268818e+05
12	furnishingstatus_unfurnished	-4.136451e+05

